

The record's distance is a path length

Three results that close STATE's next items 1 and 2 of 2026-09-23.

The single-shot constant for every protocol. Every protocol theorem of the programme — the aperture bound of the probabilistic note, the recoil bound of the recoil note, the disturbance bound of the disturbance note — holds with $\arcsin(1 - 2\epsilon)$ in place of $1 - 2\epsilon$ (Theorem P). Since $\arcsin(1 - 2\epsilon) = \arccos(2\sqrt{\epsilon(1 - \epsilon)})$, this is the sharp single-shot constant of Theorem 1 of the probabilistic note, so the gap between the single-shot and protocol forms is closed, and at certain decision the recoil bound reads

$$s \sum_j \Delta_j \geq 4\pi\hbar = 2h.$$

The aperture bound in its kick form. For every protocol of any instruments, with the same preparation under both hypotheses,

$$\hbar \arcsin(1 - 2\epsilon) \leq \int_0^\tau |f(t)| L(t) dt,$$

with $L(t)$ the body's position spread at time t (Theorem K). A force is a succession of impulses, and an impulse is registered only through the position spread it acts on. The (L, P) form of the probabilistic note follows because free motion spreads position at the rate P/m , and the two-packet preparation saturates the kick form exactly as $\epsilon \rightarrow 0$.

One accounting. The aperture bound pays for the force at the body, the disturbance bound pays for it at the marks. Any split of the force between the two gives a valid bound (Theorem M), and for von Neumann marks and a force of one sign the best split is pointwise:

$$\hbar \arcsin(1 - 2\epsilon) \leq \int_0^\tau |f(u)| \min(L(u), R(u)) du, \quad R(u) = \frac{1}{m} \sum_j \Delta_j G_\tau(t_j, u),$$

with G_τ the Dirichlet Green's function of the interval. $R(u)$ is the **recoil length**: the marks' recoil spreads weighted by the displacement each one would produce at time u relative to the chord pinned at the ends, summed, which is a length (an upper bound on the spread those recoils produce, since independent recoils add in quadrature). So the force is paid in length, at every moment the smaller of the body's spread and the marks' recoil length (Corollary M1).

Prior art. Lemma 1 is the mixed-state Mandelstam–Tamm bound in the Bures angle (Uhlmann, Phys. Lett. A **161**, 329, 1992; Anandan and Aharonov, PRL **65**, 1697, 1990); statistical distance as a path length is Wootters's (PRD **23**, 357, 1981) and Braunstein and Caves's (PRL **72**, 3439, 1994); the hybrid chain is the hybrid argument of Bennett, Bernstein, Brassard and Vazirani (SIAM J. Comput. **26**, 1510, 1997); Bures-angle speed limits for general processes are those of Taddei and coauthors (PRL **110**, 050402, 2013); and the position-spread form of force sensitivity is the waveform estimation limit of Tsang, Wiseman and Caves (PRL **106**, 090401, 2011); all metadata. New here are the split accounting of Theorem M and the recoil length of Corollary M1. Exploratory; no ledger promotion.

1. The Bures angle

For density operators ρ, σ let $F(\rho, \sigma) = \text{Tr}|\sqrt{\rho}\sqrt{\sigma}|$ be the root fidelity and $A(\rho, \sigma) = \arccos F(\rho, \sigma) \in [0, \pi/2]$ the Bures angle. Four standard facts are used.

1. A is a metric on states: it obeys the triangle inequality.
2. A does not increase under any channel, measurements and classical post-processing included (monotonicity of the fidelity).

3. $F(\rho, \sigma) = \max |\langle \psi | \phi \rangle|$ over purifications (Uhlmann, Rep. Math. Phys. **9**, 273, 1976, metadata), so $A(\rho, \sigma)$ is at most the Fubini–Study angle between any two purifications.
4. $\frac{1}{2} \|\rho - \sigma\|_1 \leq \sin A(\rho, \sigma)$ (Fuchs and van de Graaf, IEEE Trans. Inf. Theory **45**, 1216, 1999, metadata); for probability distributions this is the classical statement.

Lemma 1 (one unitary step). Let ρ be a state of body, apparatus and a classical record register x , $\rho = \sum_x p_x |x\rangle\langle x| \otimes \rho_x$, and let $V = \sum_x |x\rangle\langle x| \otimes V_x$ with $V_x = \mathcal{T} \exp(i \int_0^1 K_x(u) du)$. Then $A(\rho, V\rho V^\dagger) \leq \sup_x \int_0^1 \Delta K_x(u) du$, each spread taken in the state $V_x(u)\rho_x V_x(u)^\dagger$ along the path.

Proof. Square roots of block-diagonal operators are block diagonal, so $F(\rho, V\rho V^\dagger) = \sum_x p_x F(\rho_x, V_x \rho_x V_x^\dagger)$, and $\sum_x p_x \cos A_x \geq \cos \sup_x A_x$. For each block, purify ρ_x to ψ_x ; by fact 3 the angle is at most the Fubini–Study length of the curve $V_x(u)\psi_x$, whose speed is the spread of $K_x(u)$ in the current state, and that spread equals the spread in the mixed state it purifies. $\square$

For a time-independent generator G acting on the body, the bound is $\sup_x \Delta_{\rho_x} G$, the spread in the body’s marginal.

2. Theorem P: the single-shot constant for every protocol

Theorem P. In Theorem 2 of the probabilistic note, Theorem R and Corollary R1 of the recoil note, Theorem U and Corollary U1 of the disturbance note, and Theorems K and M below, the conclusion holds with $\arcsin(1 - 2\epsilon)$ in place of $1 - 2\epsilon$.

Proof. Each proof builds a chain of hybrid processes $H_0, \dots, H_n$ joining the two hypotheses, in which consecutive hybrids apply the same channel before and after one step where they differ by a unitary. Replace the triangle inequality for total variation by fact 1, data processing by fact 2, and the per-step trace-distance bound by Lemma 1. This gives $A(P_I, P_F) \leq \Sigma$, the same sum Σ of spreads over $\hbar$ that each proof obtains. A test with equal-prior error at most ϵ has $\frac{1}{2} \|P_I - P_F\|_1 \geq 1 - 2\epsilon$, so by fact 4 $\sin A \geq 1 - 2\epsilon$, hence $\Sigma \geq A \geq \arcsin(1 - 2\epsilon)$. In the recoil theorem the two probe families are products, and the chain replaces one factor at a time. $\square$

The constant is sharp in the single-shot and aperture forms. For a single shot it is Theorem 1 of the probabilistic note, the aperture bound’s protocol form is now the same inequality with the same constant, and the two-packet preparation attains the kick form as $\epsilon \rightarrow 0$ (§3). The recoil and disturbance bounds are not shown attained: the best known recoil protocol sits a factor of about 2.1 above the bound at five per cent error. At $\epsilon \rightarrow 0$ the constant is $\pi/2$ where it was 1.

3. Theorem K: the aperture bound in its kick form

Write $f(t)$ for the force history, F -hypothesis against free motion, with the same preparation σ under both. Let $L(t)$ be the supremum of the body’s position spread $\Delta \hat{y}$ at time t , in the laboratory frame, over the conditional states that can occur at time t when the force acts on any initial part of the interval.

Theorem K. Every protocol of instruments, of any kind, adaptive or not, deciding at error ϵ satisfies $\hbar \arcsin(1 - 2\epsilon) \leq \int_0^\tau |f(t)| L(t) dt$.

Proof. Let hybrid H_{t^*} apply the force on $[0, t^*]$ and none afterwards, the marks unchanged; H_τ is F and H_0 is I . Take a partition containing the mark times. On a cell $[t, t + \delta]$ free of marks, $H_{t+\delta}$ and H_t agree up to t and after $t + \delta$, and differ by the forced evolution over the cell, which for linear dynamics is the free evolution followed by the translation by the displacement the force produces from rest, $(\int_t^{t+\delta} (t + \delta - s) f ds / m, \int_t^{t+\delta} f ds)$. Its generator has spread at most $|\int_t^{t+\delta} f| \Delta \hat{y} + O(\delta^2) \Delta \hat{p}$ in the state it acts on. Lemma 1 and Theorem P’s chain give $\hbar A \leq \sum_{\text{cells}} |\int f| L + O(\delta) \sup \Delta \hat{p}$, and refining the partition gives the integral when the momentum spread of the chain’s states is bounded. If it is infinite for some conditional state

on some cell, free evolution makes that state's position spread infinite at every later time of the cell, so $L = \infty$ on a set of positive measure and the theorem holds vacuously. $\square$

Corollary K1 (the aperture form). If the conditional states after each mark, and at $t = 0$, have position spread at most L and momentum spread at most P , then between marks the spread grows at most linearly, $L(t) \leq L + P(t - t_j)/m$, because a c-number force translates and free motion shears. For the constant force,

$$F\tau L + \frac{F\tau^2}{2m}P \geq \hbar \arcsin(1 - 2\epsilon),$$

and with marks every δt the second term shrinks to at most $F\tau \delta t P/2m$. $\square$

This is Theorem 2 of the probabilistic note with the sharp constant, and it holds under either reading of that note's aperture. With spreads bounded in the interaction picture, $\Delta(\hat{y} - \hat{p}t/m) \leq L$, the laboratory spread is at most $L + Pt/m$, and the same integral results.

Saturation. The two-packet preparation of the probabilistic note, separation $\pi\hbar/(F\tau)$, has position spread $\pi\hbar/(2F\tau)$ up to a relative correction of order $\sqrt{\tau\Delta E/\hbar}$ from spreading. So $\int_0^\tau F L dt \rightarrow \pi\hbar/2$ as $\tau\Delta E \rightarrow 0$, while its error probability tends to zero and $\arcsin(1 - 2\epsilon) \rightarrow \pi/2$. The kick form is attained exactly in that limit.

4. Theorem M: one accounting

Theorem M. Split the force as $f = g + h$, and let c be any function on $[0, \tau]$ with $c'' = g/m$. For every protocol of instruments deciding at error ϵ whatever the body's initial state under each hypothesis (the composite requirement of Theorem U; with the same initial state assumed under both, the chain's mixed pair would only give $\arcsin(1 - 4\epsilon)$),

$$\hbar \arcsin(1 - 2\epsilon) \leq \int_0^\tau |h(t)| L(t) dt + \sum_j \sup \Delta(c(t_j)\hat{D}_j - mc'(t_j)\hat{X}_j),$$

with $\hat{D}_j, \hat{X}_j$ the disturbances of mark j as in the disturbance note, and $L(t)$ taken over the states of the chain below.

Proof. Take the pair of initial states σ under I and $T_c(0)\sigma$ under F, $T_c(t)$ the translation by $(c(t), mc'(t))$. The chain has two legs. First, starting from the process with force f from $T_c(0)\sigma$, switch off h cell by cell as in Theorem K, paying $\int |h|L/\hbar$; this ends at the process with force g from $T_c(0)\sigma$. Since $c'' = g/m$, the process with force g from $T_c(0)\sigma$ is the free process from σ translated by $T_c(t)$ between marks, which is the setting of Theorem U; its chain reaches the free process from σ and pays the second sum. Lemma 1 bounds every step and fact 1 adds the legs. $\square$

With $h = f$ and $c = 0$ it is Theorem K; with $h = 0$ and $c = P - a - bt$ it is Theorem U. The theorem reads as a transport of the offset between the hypotheses. A free line of offset is absorbed in the initial state at no cost, curvature carried through a mark costs that mark's disturbance weighted by the offset it meets, and force removed at the body costs impulse times position spread. The record's Bures distance is at most the length of any such transport, and a protocol escapes the floor only if at every moment the body is spread, or the marks disturb, enough to cover the force acting then.

Corollary M1 (the force is paid in length). For von Neumann marks with recoil spreads Δ_j and a force of one sign, take $g = \chi f$ with $0 \leq \chi \leq 1$ and c the solution vanishing at 0 and τ , $c(t) = -\frac{1}{m} \int_0^\tau G_\tau(t, u)g(u) du$, $G_\tau(t, u) = \min(t, u)(\tau - \max(t, u))/\tau$. Then $\sum_j \Delta_j |c(t_j)| = \int \chi |f|R$, and minimizing over χ pointwise, which is legitimate because for von Neumann marks the chain's conditional spreads do not depend on χ ,

$$\hbar \arcsin(1 - 2\epsilon) \leq \int_0^\tau |f(u)| \min(L(u), R(u)) du, \quad R(u) = \frac{1}{m} \sum_j \Delta_j G_\tau(t_j, u). \quad \square$$

A mark at 0 or at τ contributes nothing to R , whatever its recoil, since G_τ vanishes there; its reading still fixes the chord. For the three marks of Proposition D of the mark-cost note, outer marks sharp, $R(u) = \Delta_2 \min(u, \tau - u)/2m$, a tent of height $\Delta_2 \tau/4m$. The pure recoil bound is $F \int R = \Delta_2 s/4$; the pure aperture bound is $F \int L$. The recoil length vanishes at the ends and peaks in the middle, so the mixed bound pays at the marks near the ends and at the body wherever the body is more compact than R . Whenever L and R cross, $\int F \min(L, R)$ is strictly below both pure costs, and the necessary condition it states is strictly stronger than either.

5. The numbers

Bound	Old constant	Sharp constant	At $\epsilon = 0.05$	Certain decision
Recoil, $s \sum_j \Delta_j / \hbar \geq$	$8(1 - 2\epsilon)$	$8 \arcsin(1 - 2\epsilon)$	8.96	4π , i.e. $s \sum \Delta_j \geq 2\hbar$
Disturbance, $(\frac{s}{8} \sum \Delta(\hat{D}_j) +$ $\frac{J}{2} \sum \Delta(\hat{X}_j)) / \hbar \geq$	$1 - 2\epsilon$	$\arcsin(1 - 2\epsilon)$	1.12	$\pi/2$
Aperture, $(F\tau L +$ $\frac{F\tau^2}{2m} P) / \hbar \geq$	$1 - 2\epsilon$	$\arcsin(1 - 2\epsilon)$	1.12	$\pi/2$

The balanced aperture of minimal area now gives $\tau \Delta E \geq \frac{1}{2} \hbar \arcsin^2(1 - 2\epsilon)$, which is $\pi^2 \hbar/8$ at certain decision. The correlated three-mark protocol sits within $\sqrt{2} z_{1-\epsilon} / \arcsin(1 - 2\epsilon)$ of the recoil bound, about 2.1 at five per cent error, at every squeezing.

6. Consequence for STATE

Items 1 and 2 of STATE's queue are settled, apart from the worst-case interval [9, 36] of the derivation note, which concerns a framework without quantum instances. Every protocol theorem carries the sharp single-shot constant; the aperture bound is the kick form $\int |f| L \geq \hbar \arcsin(1 - 2\epsilon)$; and one accounting, Theorem M, contains the aperture and disturbance bounds and interpolates between them, with the pointwise form $\int |f| \min(L, R)$ for von Neumann marks. The remaining modern item is whether the minimum over splits and lines in Theorem M is attained by some protocol, which would make it the exact floor.